

A Framework of AI Strategy

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Abstract

Building a vibrant strategy for the use of Artificial Intelligence (AI) in business is becoming critical for company success in the AI era. To decode AI strategy, this study aims to address three questions: (1) What is AI strategy? (2) What are the key elements of AI strategy? and (3) How can AI strategy be applied and impact a business? We define AI strategy and propose an AI strategy framework consisting of three layers: AI strategic competency, AI use cases, and AI enabling factors. Based on published cases that describe early applications of AI, we identify a bottom-up approach typically used by Small and Medium-sized Enterprises (SMEs) in building AI strategy and a top-down approach typically used by big tech firms and traditional incumbent firms. We also find differences in the core of the AI strategies that are initiated by the three types of enterprises: SMEs: survival oriented; big tech firms: AI ecosystem oriented; traditional incumbent firms: business alignment oriented.

Keywords: AI, AI Strategy, Competitive Advantage, Digital Transformation.

1. Introduction

Artificial Intelligence (AI) is becoming a critical element for the competitiveness, sustainable development, and overall success of companies because of its incredible recent advances and ubiquitous applicability in business and society (Di Vaio et al., 2020; Krakowski et al., 2023; Mahidhar & Davenport, 2018). AI is used to enhance the experience of customers by intelligently processing vast amounts of customer data (Huang & Rust, 2022), to manage repetitive and routine knowledge work tasks (Davenport & Ronanki, 2018), to leverage strategy and optimize the measurement of key performance indicators in organizations (Kiron & Schrage, 2019), and to handle many more tasks that have not been explored yet. The transition from human-oriented tasks and decisions to

AI-assisted tasks and hybrid decisions results in changes to business models, organizational structures, and, in turn, changes to the value propositions of organizations (Dwivedi et al., 2021). Pioneer organizations in private and public sectors and even national governments have started to develop strategies for adopting, advancing, and applying AI technology to create opportunities and value throughout their business activities (Joshi & Wade, 2020).

Although the importance of AI has been well recognized, a great number of studies focus on technical questions about developing or applying AI but there is a lack of studies that explore AI with respect to strategic management (Berente et al., 2021; Keding, 2021). AI applications reflect the heterogeneity of the scenarios in which they are utilized and have started to penetrate many facets of business operations. However, common guidelines for strategy building are still missing. With this study, we intend to move closer to providing an overarching and systematic guide for enterprises seeking to build a vibrant AI strategy. Specifically, we focus on three questions: (1) What is AI strategy? (2) What are the key elements of AI strategy? and (3) How can AI strategy be applied and impact the business?

To answer those questions, we review previous studies that apply strategic views of technology and that can be applied to AI. IS strategy has been defined as “the organizational perspective on the investment in, deployment, use, and management of information systems” (Chen et al., 2010, p. 237). Applying Chen et al.’s (2010) notion to AI, we define AI strategy as the *organizational perspective on the investment in, deployment, use, and management of AI technology and AI systems*. In addition, a shared perspective (Mintzberg, 1987) on AI strategy and its components is necessary to guide the selection of development and adoption alternatives, the acquisition and preparation of data sources (Correani et al., 2020), training and reconfiguration of models, and the maintenance and fine-tuning of AI infrastructure and services. A shared perspective of AI refers to a common view of a given

organizational locus and direction in a particular area which is communicated clearly with all relevant stakeholders. Nowadays, the development of AI pipelines is time consuming, meaning that wait-and-see approaches can be problematic in a fast-paced environment (Mahidhar & Davenport, 2018). Formulating an AI strategy enables a learn-fast and fail-fast approach instead. Building on prior literature, we refine our notion of AI strategy with a focus on both existing AI operations as well as future initiations.

Our study is structured as follows: We start with a literature review in section 2, then present a framework for AI strategy that integrates the core elements identified in the literature review (section 3). The framework combines elements in three layers. The top layer, AI strategic competency, includes AI vision, competitive advantage, and customer value. The middle layer, AI use cases, includes AI-powered products and services and AI-powered operations. The bottom layer, AI enabling factors, includes people, organization, technology, and AI ecosystem. In section 4, we examine published case studies of AI applications and identify two approaches for building an AI strategy: the bottom-up approach and the top-down approach. While the former method is used primarily by Small and Medium-sized Enterprises (SMEs), the latter method is used more often by big tech firms and traditional incumbent firms. In addition, we identify differences in the core of the AI strategies that are initiated by the three types of enterprises. SMEs' AI strategies are survival-oriented, big tech firms' AI strategies are AI ecosystem-oriented, and traditional incumbent firms' AI strategies are business alignment-oriented. Section 5 concludes the paper.

2. Literature Review

2.1. Strategy and IS Strategy

Strategy refers to a long-term plan for how a business is going to compete, what its goals should be, and what activities the company will perform to carry out those goals (Mintzberg, 1987). Ideally in line with a higher-level and stable vision for the organization (Collins & Porras, 1996), a good strategy can allow a company to outperform its rivals by delivering a unique mix of values (Porter, 1996). As a strategy is implemented, it informs decisions about resource allocation, operations, and specific projects. While strategies have long been implemented in a somewhat static fashion to cover a period of about three to five years, the fast-paced developments of contemporary business environments have resulted in more dynamic, rolling approaches to strategy (Teece et al., 2020).

From a practical perspective, strategy can be developed not only for an entire organization, but also for any of its units, sub-units, and functions across the organizational hierarchy, including information systems (IS) as a whole or for parts of it. Maintaining consistency among the various strategic plans is important to avoid conflicts and inefficiencies.

At the organizational level, Chen et al. (2010) describe the following three concepts of IS strategy: “(1) IS strategy as the *use of IS to support business strategy*; (2) IS strategy as the *master plan of the IS function*; and (3) IS strategy as the *shared view of the IS role within the organization* (p. 238, emphases in the text). The third conception, IS strategy as a shared view of the IS role within the organization is suggested to be most appropriate to be applied in the context of building a strategy for impactful innovations because it best balances the perspectives of the business and IS related to the technology (Chen et al., 2010).

Correani et al. (2020) emphasize the need to properly execute a strategy to achieve its value. Following the definition of a “guiding policy for the creation and appropriation of value by exploiting digital technologies to achieve long-term objectives” (i.e., strategy formulation), firms need to “translate the digital strategy formulated into a concrete plan and set of actions” (p. 38). To achieve consistency between the actions and objectives, Correani et al. (2020) highlight the four building blocks of resources, capabilities, activities, and stakeholders and emphasize shifts that may happen in relation to technology innovations. For example, the role of data might be more central in the recent context of digital transformation and AI than more conventional business models of the past.

2.2. AI and AI Strategy

In the following, we first point out previous work on the particularities of modern AI, before we review studies that have suggested the need for comprehensive and long-term AI strategies and that have provided starting points for AI strategy development.

AI is commonly associated with the capability of machines to emulate certain traits of human intelligence (Benbya et al., 2021; Russell & Norvig, 2020). While AI has been around for many decades, recent advances in computing power and algorithms, as well as the availability of large volumes of high-quality data and powerful, typically cloud-based, storage solutions have accelerated its commercialization and establishment in business operations and everyday lives (Benbya et al., 2021; Dwivedi et al., 2021).

Berente et al. (2021) emphasize the dynamic aspect of AI as a moving target when they refer to it as “the frontier of computational advancements that references

human intelligence in addressing ever more complex decision-making problems” (p. 1435). Contemporary forms of AI are characterized by the three interrelated facets of autonomy, learning, and inscrutability (Berente et al., 2021). Autonomy means that AI-systems can act on their own and make autonomous decisions, including generative behavior and new ways of interacting with human and artificial agents, all with possibly profound implications for management and knowledge work. Learning refers to the ability of AI-systems to improve through data and experience inductively and automatically. Learning includes new methods of deep and reinforcement learning that utilize ever more powerful computing and new forms and very large quantities of data within and beyond the confines of the organization. AI-powered systems can be capable of generating new knowledge with limited human oversight or input. Inscrutability refers to the use of algorithmic models that are intelligible to at most very few humans, characterized by varying levels of opacity, transparency, explainability, and interpretability with implications for the “black box” phenomenon often attributed to AI and other management issues.

Among the different forms of contemporary AI (e.g., robotics, expert systems, neural networks), machine learning (ML) has been labeled as the most relevant and lucrative (Benbya et al., 2021; Hartman, 2023; Jacobides, 2021). ML “uses statistics to find patterns in large amounts of data and uses these patterns to make predictions” (Hartman, 2023, p. 2). Recent applications of AI can provide recommendations and increasingly automated support for complex decisions in areas, such as knowledge work in banking, insurance, customer relationship management, drug discovery, toxicology, or bioinformatics (Benbya et al., 2021). The discussion about to what extent and how AI substitutes (automates) or complements (augments) human intelligence is ongoing (Krakowski et al., 2023).

Important components of modern AI systems relate to data management (sourcing and integration of large volumes of high-quality data), system and model development and refinement, and deployment infrastructure, including computation and storage (Benbya et al., 2021; Jacobides et al., 2021). Interestingly, the developers and providers of critical AI components, such as Google, Amazon, Microsoft, and other members of “big tech”, are also among AI’s biggest users and benefactors (Jacobides et al., 2021).

We also note that AI is conceptually close to digital transformation because it enables and supports the use and interpretation of large amounts of data that drive digital transformation (Gurbaxani & Dunkle, 2019). Consequently, AI can play a role in taking digital transformation to the next level (Correani et al., 2020).

Mahidhar and Davenport (2018) point out that adopting and fully realizing the value of AI can take longer than for regular IT, thus requiring more upfront planning (i.e., strategizing) for several reasons. AI systems often need to be configured to fit an organization’s unique situation. Then they also need to be trained with data that may first need to be collected and prepared. For areas where knowledge domains have not yet been modeled by a vendor, a knowledge architecture may have to be set up and configured. AI must be integrated into an organization’s processes. And it needs to be decided what implications the AI systems have for the workforce. All these aspects increase the urgency of developing an AI-related strategy sooner, rather than later.

Joshi and Wade (2020) indicate that many companies use AI mainly for “tactical rather than strategic purposes” and a “cohesive long-term AI strategic vision is rare” (Joshi & Wade, 2020). To address the lack of longer-term strategic vision and support the development of a sustainable approach to AI, Joshi and Wade (2020) suggest building an AI strategy on three pillars: (1) robust and reliable technology infrastructure, in particular as it relates to usable data and feedback loops; (2) new AI-enabled business models; and (3) AI-related ethics to counter AI-related bias and ensure accountability.

Li et al. (2021) define AI orientation, which corresponds with our notion of AI strategy, as “a firm’s overall strategic direction and goals associated with introducing and applying AI technology,” which “provides managers with a framework to guide how they invest, manage, and apply AI within their firms” (p. 1604). The underlying goal of developing an AI orientation is to “maximize the benefits of AI while minimizing the associated costs and risks” (p. 1604). It is intended to help managers make decisions regarding investments, applications, and communication with stakeholders. The need for AI-related strategy results from the specific characteristics of the technology but also from its novelty and the many associated unknowns. Aspects to be included in strategies related to current developments of AI include not only the provisioning of adequate technology and systems that are sufficiently powerful and scalable, but also objectives, guidelines, policies, and standards to assess the potential for AI-based and -supported applications and their risks and limitations, determine specific uses, evaluate the outcomes of AI-based decisions and AI-generated content, and manage the implications of AI in general.

Kiron and Schrage (2019) suggest using key performance indicators (KPIs) to define a company’s AI strategy that then provides corporate leaders with a basis to create value, accountability, and competitive

advantage. They also emphasize the need to develop a strategy not just *for* AI but also *with* AI and state that “optimizing carefully selected KPIs becomes AI’s strategic purpose” (p. 31). For example, ML could be applied to choose, measure, and optimize the organization’s selected KPIs, whereby data and careful data management again play an important role to enhance the measures and help the machines learn over time.

Krakowski et al. (2023) argue that AI as a potential supplement and complement of human intelligence alters the sources of competitive advantage. Managers have “to develop new capabilities to stay relevant in an AI-based competitive landscape” (p. 1426). AI can *substitute* domain-specific capabilities, e.g., related to managerial decision making, if it helps non-experts perform better and consequently erodes the competitive advantage of the domain experts. At the same time, there can also be a *complementary* effect when the technology (which is general-purpose to some extent) is used to develop unique bundles of competencies that augment the domain knowledge. Competitive advantage can consequently shift to those who best understand what to do with and how to apply the technology, which might be associated more with capabilities related to the technology (e.g., AI literacy) or creativity than the original domain.

A good AI-related strategy will understand the potential of AI to substitute traditional capabilities, e.g., to make managerial decisions and solve business problems and ensure that the organization stays in play as the technology is increasingly becoming a business necessity. At the same time, a good AI-related strategy should also consider ways to augment human expertise with the goal of developing unique capabilities that have the potential to afford new competitive advantages.

3. AI Strategy Framework

Under the umbrella of the strategy definition, it is suggested that AI strategy is a part of a company’s business strategy and refers to a plan of action for how an organization intends to strategically build and leverage AI technologies to achieve its goals and objectives (Hartman et al., 2023). A well-defined AI strategy can help an organization align its AI initiatives with the overall business strategy and to prioritize and allocate resources to the most promising AI initiatives.

To build a conceptual framework, we conducted a literature review with the commonly accepted guidelines (Rowe, 2014; Webster & Watson, 2002). First, we performed a literature search in the EBSCOhost database (Business Source Ultimate) with the search terms of “AI” and “artificial intelligence” in combination with “strategy”. For the sake of timeliness,

relevance, and quality, we limited the search results to the articles published in top journals in IS (Decision Support Systems, European Journal of Information Systems, Information & Management, Information and Organization, Information Systems Journal, Information Systems Research, Journal of Information Technology, Journal of Management Information Systems, Journal of Strategic Information Systems, Journal of the Association of Information Systems, MIS Quarterly) and management (Academy of Management Journal, Academy of Management Review, Decision Sciences, Harvard Business Review, Management Science, Strategic Management Journal). The database search yielded 431 articles. In addition, we applied the same search teams and relevant topics in Google Scholar. After reviewing the titles, abstracts, and full-texts of the articles, we excluded duplicates and articles of no and low-relevancy to AI strategy. The screening process yielded 29 articles (12 from EBSCOhost; 17 from google scholar) as the final relevant set of studies.

Based on the identified relevant studies (e.g., Berente et al., 2021; Bharadwaj, 2000; Floridi et al., 2018; Hartman et al., 2023; Huang & Rust, 2022; Jacobides et al., 2021; Joshi & Wade, 2020; Keding, 2021; Kemp, 2023; Krakowski et al., 2023; Raisch & Krakowski, 2021; Ravichandran, 2005), we propose an AI strategy framework that consists three layers, each comprising of several elements: AI strategic competency, AI use cases, and AI enabling factors, (Figure 1).

3.1. Top Layer: AI Strategic Competency

3.1.1. AI Vision. AI vision constitutes the top layer of the framework. On the one hand, AI vision must be under the arch of the business strategy which is not included in the framework to align with the big picture of the company. On the other hand, AI vision directly and indirectly reconciles all the elements in the framework. At the top layer, two important aspects derived from AI vision are competitive advantage and customer value (Collins & Porras, 1996). The stakeholders of AI strategy need to figure out: (1) What is the situation of the company in the competition? (2) Who are the existing and potential competitors before and after adopting AI as part of the company’s business? (3) What is the real value that the company can provide to customers through AI?

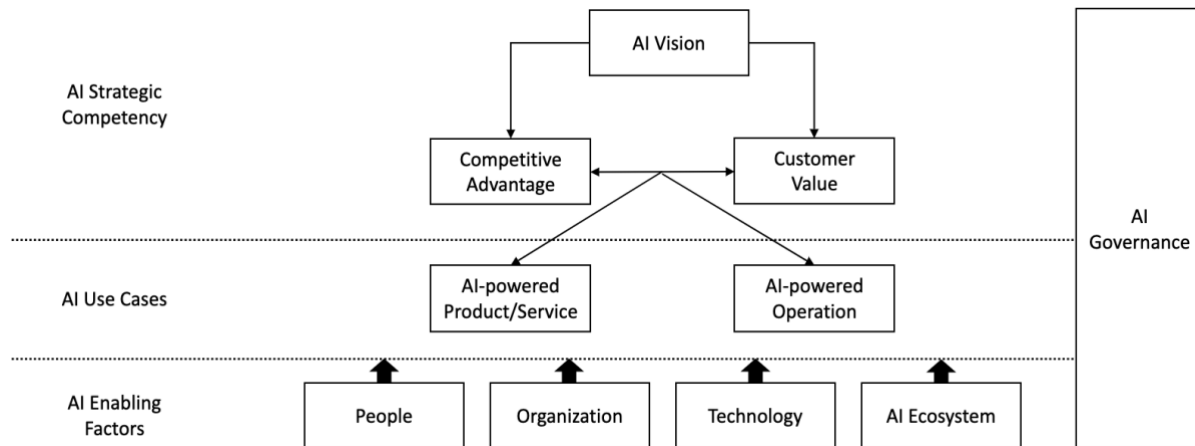


Figure 1. AI strategy framework

3.1.2. Competitive Advantage. Gaining competitive advantage through AI needs to deal with three strategic limitations of AI: the generic, explicit, and myopic nature of AI, which have critical adverse effects (Kemp, 2023). AI is built primarily on algorithms, data, IT infrastructure, and computing devices (Jacobides, 2021). Since hardware like IT infrastructure and computing devices can be easily acquired in the market, algorithms and data become the critical factors to create unique capabilities. However, many algorithms for building AI features such as natural language processing are generic, which can mean similar results for companies using the same algorithms. For example, “all else being equal, a neural network algorithm will arrive at the same logic for connecting a set of inputs to outputs regardless of whether Spotify or Pandora operationalizes the neural network” (Kemp, 2023, p. 7). In addition, the basic element of algorithms is code which is explicit and can be replicated by competitors. Last, AI is good at handling tasks given a specific scope but lacks contextual awareness without a clear boundary such as a firm’s strategy. Therefore, designing unique AI use cases in the middle layer, capturing unique data, and synthesizing the internal technologies as well as other resources at the bottom layer for training AI are critical steps to building competitive advantages.

3.1.3. Customer Value. AI techniques have been used to provide value to customers in many areas. Although the resulting product or service value does not always signal the use of AI, the technology can provide a smooth, easy, or seamless user experience (Davenport, 2018; Di Vaio et al., 2020; Huang & Rust, 2022). The customer value created through AI typically results either from improving work efficiency and in consequence lowering cost or from creating unique

products, services, or experiences, or both. At this point, it is important for AI strategy makers to define their position with respect to customer value. The typical formats of AI applications are depicted in the middle layers.

3.2. Middle Layer: AI Use Cases

Linked to the top layer, the AI use case layer defines how to provide value to users. As AI enables machines and systems to perform tasks like a human and to become more intelligent, the range of applications and use cases increases. The strategic value of AI manifests itself in the form of products, services, and operations that benefit from AI at some level as AI provides augmentation or automation or both (Raisch & Krakowski, 2021). In general, we classify AI use cases into two categories: AI-powered product and service, and AI-powered operation.

3.2.1. AI-powered Product and Service. Many organizations have designed new products and services or improved already existing ones through the power of AI to gain competitive advantage and deliver value for their customers (Kemp, 2023). We are seeing many cases across industries. For instance, AI-enabled chatbots can provide 24/7 customer service, handle frequently asked questions, and support customers in real-time (Davenport, 2018). ByteDance which developed TikTok has gained significant competitive advantage among social platforms embedded with AI through providing value to the content creators through advanced content-recommendation (Davis et al., 2019). C3 develops products and services for the Air Force that leverage services provided by cloud AI platforms to

create systems for preventative maintenance of mechanical systems (Simon & Gonzalez, 2018).

3.2.2. AI-powered Operation. The management literature has identified two broad areas of AI applications: (1) automation applications where the tasks and processes are completed by the machines only; (2) augmentation where tasks and processes completed by humans are improved by machines (Brynjolfsson & McAfee, 2014; Davenport & Kirby, 2016; Dougherty & Wilson, 2018). However, Raisch & Krakowski (2021), argue that although the automation-augmentation distinction is intuitive, it overlooks the interdependencies among the two categories of applications and that their boundaries are blurred across time and space. The augmentation versus automation is sometimes also termed complementation versus substitution through AI (Krakowski et al., 2023) and may be a valid starting point for the formulation of an AI strategy. FireEye, for instance, applies AI to detect suspicious network activities but also relies on human experts to identify new potential cyber-attack scenarios (Miller & Bhattacharya, 2021).

3.3. Bottom Layer: AI Enabling Factors

AI enabling factors constitute the bottom layer in the framework and provide the foundation for AI strategy and implementation. Although many factors may impact the success of an AI project, people, organization, technology, and the AI ecosystem are identified as indispensable elements supporting company growth in many studies (Bharadwaj, 2000; Hartmann et al., 2023; Jacobides et al., 2021; Ravichandran et al., 2005).

3.3.1. People. People constitute a critical part of any project and the related changes. Organizational human resources significantly impact the progress and result of an AI project. People include both new management roles that drive AI projects and the employees who possess the skills and capabilities associated with developing and applying or interacting with AI. Companies planning to step into the AI area report challenges related to the review of in-house AI skills and capabilities, training current employees, and hiring and retaining AI experts. Furthermore, establishing and fostering trust between human operators and the AI tools they interact with is essential.

3.3.2. Organization. AI is a powerful technology that has the potential to change organizational structures and governance in many ways and transform individuals' positions in organizations. The changes can create concerns for employees that would lead them to raise

questions about the impact on organizational business processes. In some cases, there are change management and internal conflicts related to the adoption and implementation of AI. For example, PittaRosso, the Italian shoe retailer, faced internal conflicts between finance and sales when it adopted and implemented an AI-based pricing and promotion system (Israel, 2021). To orchestrate the interaction with other enabling factors and the upper layers of the framework, the stakeholders of AI strategy need to address questions related to organizational structure and governance, such as: what is the key support the AI project can expect from the organizational change? which department/business units will be impacted by the AI project? how will AI change the culture in the company? how should the change process be managed?

3.3.3. Technology. Technology constitutes another key enabling factor. The IS infrastructure and data are two primary facets of technology in the framework. AI relies heavily not only on powerful computational devices and huge amounts of data needed to train AI models, but also on reliable and fast networks that transmit data among numerous digital devices. Therefore, stakeholders of AI strategy need to understand available technology resources in the organization. More importantly, they must be clear on what technology resources are lacking that hinder realizing the AI project.

3.3.4. AI Ecosystem. The AI ecosystem interconnects other enabling factors in the framework. Most companies do not have sufficient resources in the form of people, data, or infrastructure to develop an AI project on their own. An AI ecosystem can enable industries where companies want to collaborate with partners who possess the resources or techniques that they need (Jacobides et al., 2021). This channel can ease the challenges associated with two critical AI enabling factors: people and technology.

Jacobides et al. (2022) identified three distinct areas in an AI ecosystem: AI enablement, AI production, and AI consumptions. The ecosystem involves developers, manufactures, and users of AI. Depending on the core business of an organization and the AI use cases, any or a combination of these may characterize the role of the organization in the ecosystem. In the context of AI strategy within a single firm, the AI ecosystem refers those factors that motivate, enable, mandate, or make feasible the aspiration of the organizations for AI-related innovations in products, services, and operations. The ecosystem involves entities such as hardware manufacturers like NVIDIA (Watkins et al., 2019) and platform providers like AWS or Numenta that enable AI products provided by downstream organizations such as C3 (Simon & Gonzalez, 2018) as

well as suppliers, partners, or regulatory bodies that demand an advancement on the augmentation or automation scale of the company.

3.4. AI Governance

As AI, to some extent, has an ability of sensing the environment and is being trained to think and behave like a human, a concern might arise in cases where the output generated by an AI system cannot be fully controlled (Berente et al., 2021) and can be associated with all the elements from the three layers. Therefore, we suggest regarding AI governance as a more general component that transcends the other layers.

Many test results of AI applications might be considered misleading, biased, and even dangerous. They are not ethical or adhere to regulation, law, and convention (Benjamin, 2019; Jobin, 2019). For instance, ChatGPT is a chatbot which interacts in a conversational way and has quickly gained attention for its detailed responses and articulate answers across many domains of knowledge. However, its uneven factual accuracy was identified as a significant drawback. People have used ChatGPT to flood Stack Overflow, the go-to question-and-answer site for coders and programmers, with AI responses that seem correct at first glance but are often wrong on close examination. The site has temporarily banned users from sharing responses generated by ChatGPT.

Related to AI applications in management, assessing the consequences, evaluation equity, privacy, and transparency (Castelvecchi, 2016) of impacts are pressing issues. Only a fraction of organizations has started to explicitly address ethical governance. Mastercard, for instance, has leveraged its existing privacy framework to include assessment, mitigation, and auditing of AI-related processes (Isik & Duke, 2022).

AI governance plays an important role across the layers of our AI strategy framework because the underlying issues are complex and often rooted in multiple perspectives. Key questions to ask include: How can we avoid invading people's privacy? Are there any legal implications of using AI in this way? What sort of consent do we need from customers/users/employees? How can we ensure our AI is free of bias and discrimination?

4. AI Strategy Framework Applications

The AI strategy framework is intended as a guide for companies who plan to or are on the trajectory to adopt AI in their business. To assess the applicability of the framework, we examined 65 cases out of 417 searched articles, book chapters, case studies, and

industry background notes that we retrieved from the Harvard Business Publishing Education database (<https://hbsp.harvard.edu/>) using the keyword "AI Strategy". The examined cases demonstrate a variety of strategic roles that AI plays as it is adopted in business and the implications it has for business transformation.

Based on the cases that we reviewed, we find the AI strategy framework to be applicable, as AI-related strategies are described explicitly or implicitly in the papers. However, we also noticed that the AI strategy framework is not a one-size-fits-all framework. It plays out differently in different types of companies to accommodate complex situations.

4.1. Cases in Small and Medium-sized Enterprises (SMEs)

Small and midsize enterprises (SMEs) are referred to as businesses with revenues, assets, or number of employees below a certain threshold. According to the standards of U.S. Small Business Administration (2023) and considering that most AI companies fall into the information industry, we classify SMEs as companies with less than 1,500 employees.

SMEs, including startups, usually initiate AI strategy from the enabling factors and the corresponding AI use cases in the AI strategy framework. As the AI technology becomes mature and can be applied to a larger scale of customers and users, the higher level elements of the AI strategy framework, such as AI vision (e.g., what is the company's vision on adopting AI in its business?) and AI governance (e.g., how to deal with potential ethical issues or conflicts in AI application?), must be taken into greater consideration to help ensure that a vibrant AI strategy can guide the AI application as part of the business. We call this approach to building an AI strategy Bottom-Up Initiation (BUI).

Most BUI occurs in SMEs because they are small and possess fewer resources that are crucial for survival than large companies. Especially in startups, the core assets tend to be the founders and employees who are experts in certain areas of AI. SMEs often provide unique AI products or services (e.g., ChatGPT; vision recognition for self-driving vehicles). They focus on niche markets that are less likely covered by incumbent or large size companies. Given the characteristics of SMEs and relatively lower competition, this type of companies tends to start forming a systematic AI strategy from the bottom of the framework (i.e., AI enabling factors). Those crucial resources determine how the SMEs serve their customers and directly impact the survival in the market. In contrast, the higher-level elements of the framework, i.e., AI vision, competitive advantage, customer value, as well as AI governance,

tend to not directly influence survival in the short term. Therefore, the higher-level perspectives of AI strategy are often built and reshaped at later stages. Therefore, we call this type of AI strategy “survival oriented.”

4.2. Cases in Large Enterprises

To distinguish large enterprises from SMEs, we classify large enterprises as companies with more than about 1,500 employees. In contrast to the BUI approach that is typical for SMEs, large enterprises often start building AI strategies from the top of the AI strategy framework (i.e., AI vision, competitive advantage, customer value) and define the direction of AI adoption at a higher level for the organizations. During the strategy making process, large enterprises tend to be more rigorous and cautious in shifting the direction of their mature businesses and adopting AI into their organizations. Like the waterfall method in software development, this methodical, step-by-step approach tries to anticipate and minimize mistakes that may jeopardize existing businesses and operations. We label this approach to building an AI strategy Top-Down Initiation (TDI).

Based on our review of published AI-related case studies, we identified two types of large enterprises: big tech firms and traditional incumbent firms. Both possess mature resources and customers, are dominant in their industries, and tend to have comparatively mature business strategies, products, processes, organizational structures, and personnel. In the short term, they don't have survival threats. While TDI appears to be the method used by both types of big enterprises in building their AI strategy, we also found that big tech firms tend to focus more on maximizing value from building an AI ecosystem (“AI ecosystem oriented”), whereas traditional incumbent firms tend to focus more on maximizing value from aligning AI to existing businesses (“business alignment oriented”).

Big tech firms are large enterprises with primary business activities in the information and high technology industry, such as Google, Amazon, and Microsoft. They are the key participants in the AI ecosystem, as they control or have penetrated most or all stacks of AI enablement (e.g., data management/processing, computing, infrastructure) and AI production (e.g., machine learning/data analysis platform, visualization) (Jacobides et al., 2021). Relying on the foundations, which requires significant and long-term investments, the big tech firms not only create AI for internal use but also develop products, services, and technologies that offer AI affordances to other companies. Google's AutoML, for instance, allows customer organizations to employ Google services for underlying data acquisition and management, and pre-

built models to be used for organization-specific services. By providing customer value to other companies, these large players are also trying to stay competitive and capture market shares. Hence, some of the cases we reviewed reflect more than one area in the AI strategy framework. In the context of developing AI-based competencies, organizations provide programs for intelligent marketing, human-resource development, manufacturing, and business operations management. Intel Corporation's AI for Youth (Unruh, 2021) exemplifies a multinational organization investing in AI for human resource development to ensure that manufacturing and operations needs are met with local talents.

Traditional incumbent firms are large enterprises with primary business activities outside of the information and high technology sectors. They usually have dominant positions in traditional industries, such as retail, accommodation, and finance. Traditional incumbent firms tend to lack specific experience and resources for developing and applying AI. The primary concern related to the utilization of AI is to explore new opportunities that can serve the current business. Therefore, for incumbent firms, the process of AI strategy development should begin at the top of the framework. It is critical to align the AI vision with the current business vision of the company (Keding, 2021), and use the vision to link and guide the components in the middle and bottom layers of the framework. Questions to be addressed include: How to embed AI technology into current products? How to redesign business processes and revenue models to accommodate AI? and How to allocate current resources for AI development and application? Examples of this type of effort are the Hotel Innovation Incubator at Charlotte Marriott City called M Beta intended for identifying and testing strategically aligned customer-facing innovations and the Global Creative Intelligence at McCormick and Company (Meister & Chandrasekhar, 2021). Allianz used data and machine learning to alleviate plummeting sales and predict direct debits (Heyvaert et al., 2021; Van der Shraalen et al., 2022). Investment firm Vanguard has started a new “Personal Advisor Service”, which combines AI technology with investment advising that was traditionally provided primarily by human advisors. In the redesigned service, AI replaces human advisors to perform the traditional advising tasks such as customizing investment portfolios. Human advisors play the role of coaches to answer customer questions and keep customers on the right track. This redesigned service stands out from purely human advising services. It has significantly reduced the cost for Vanguard and helped customers increase their assets (Davenport & Ronanki, 2018).

5. Implications and Limitations

In this paper, we define AI strategy and propose an AI strategy framework to enhance the theoretical foundation of AI governance. To provide practical implications for top executives, AI professionals, and researchers in the area, we analyze published case studies of AI strategy applications. The framework provides a tool for organizations to identify and describe their current state with respect to the core competencies, resources, and organizational vision for the future. This understanding of the current state of AI can help articulate desired future states and potential required changes.

We acknowledge limitations of our study. For example, we have relied on published articles and case studies to build the framework and explore the patterns. The descriptions found in the cases may not fully reflect the whole picture of AI strategy across companies and industries. Future research is needed to explore the topic further, including quantitative research and natural experiments to examine the impact of the AI framework and to develop a deeper understanding about the causality between its various components.

6. References

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